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Spatiotemporal volume saliency

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Abstract This paper proposes spatiotemporal volume saliency to detect and explore salient regions in time-varying volume data. Based on the center-surround hypothesis that the salient region stands out from its surroundings, we extend the spatial saliency to time domain and introduce temporal volume saliency. It is defined as a center-surround operator on Gaussian-weighted mean attribute gradient between steps in a scale-independent manner. By combining spatial saliency and temporal saliency together, our spatiotemporal volume saliency is effective in detecting changes of salient regions. We demonstrate its utility in this regard by automating transfer function design and selecting key frames for time-varying volume data.

Keywords Volume exploration · Time-varying data · Human perception · Spatiotemporal volume saliency

1 Introduction

Time-varying volume exploratory visualization has been used to analyze time-dependent simulation from many physical and chemical processes. The rapid growth in the number and quality of simulation and the requirement of human-centered visualization suggest the need for incorporating insights from human perception into time-varying volume exploration. As a crucial part in time-varying volume exploration, transfer function assigns optical properties, such as color and opacity, to the data being visualized determining which structures of the volume will be visible and how they will be rendered. Although researchers have extensively studied transfer function design (Pfister et al. 2001), designing proper transfer functions for time-varying volume data is still a challenge (Ma 2003). Besides transfer function design, dynamic view selection (Ji and Shen 2006) is also a challenging problem. To facilitate quick exploration of the data with long time steps, clustering time steps into time intervals and providing cluster-based exploration scheme are indispensable.

Inspired by center-surround hypothesis that the salient region stands out from its surroundings (Koch and Poggio 1999), we propose spatiotemporal volume saliency, a measure of regional importance, to facilitate the time-varying data exploration. This measure consists of two parts: spatial and temporal saliency. The spatial saliency is able to capture prominent volume structures as the most salient parts of a volume, while temporal saliency can reveal significant changes of different time steps. The temporal saliency is achieved by extending spatial volume saliency (Kim and Varshney 2006) into time domain, which is defined as a center-surround operator on Gaussian-weighted mean attribute gradient between steps in a scale-

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independent manner. With that, our spatiotemporal saliency is very effective in detecting the change of salient regions.

Our spatiotemporal saliency provides an intuitive method for visual analysis of time-varying volume data, which consists of regular, periodic, and random/hot spot behaviors (Jankun-Kelly and Ma 2001). We can automate transfer function design by setting opacity based on spatiotemporal saliency. Finally, we compute the salience distance between any pair of volumes and cluster the salience distance matrix to decompose the time series into different time intervals. Then, we detect key frame in each time interval and show them to the user. With this, we provide a way to automatically design viewpoints for different time steps.

The main contributions of this paper are given as follows:

1. *Spatiotemporal volume salience* We propose spatiotemporal saliency for time-varying data and demonstrate that it is effective in detecting the changes of salient regions.
2. *Saliency-based time-varying volume exploration* We provide several saliency-based applications to help the user explore the time-varying data, including automatic transfer function design and key frame selection.

2 Related works

In this section, we review related work concerning saliency in computer graphics and computer vision. Since our saliency is used to analyze time-varying flow, we also present different strategies that have been proposed in time-varying visualization for transfer function design.

2.1 Saliency

The saliency of an item, such as pixel, is the state or quality by which it stands out relative to its neighbors. Saliency detection is considered a key attentional mechanism that facilitates learning and survival by enabling organisms to focus their limited perceptual and cognitive resources on the most pertinent subset of the available sensory data (Zhao and Koch 2013). Saliency detection has been studied in computer vision for years (Koch and Ullman 1987; Koch and Poggio 1999). Saliency is recently applied to computer graphics (Lee et al. 2005) and visualization (Kim and Varshney 2006; Yu et al. 2012). Next, we discuss saliency-related visualization work in detail.

Kim and Varshney (2006) introduced a saliency-based enhancement operator to guide visual attention in volume visualization and augmented the existing visualization pipeline by incorporating enhancement operators to increase the visual saliency of different regions of a volume dataset. However, they compute saliency field with data value only. While their method deals with the problem that highlights features near user's interaction with spherical regions of interest and binary-valued saliency field, our method handles the problem that inferring picking position with the defined saliency.

Yu et al. (2012) proposed flow saliency to assist streamline seed selection. The generated set of streamlines captures important flow features near critical points without enforcing the dense seeding condition. The defined flow saliency integrates curvature and torsion based on the Gaussian-weighted center-surrounding (Itti et al. 1998) evaluation. However, this method only considers the geometry of streamline without other useful information. As a result, it could not capture region with small-scale shape-changing streamlines.

Saliency could be also used to evaluate the quality of visualization (Janicke and Chen 2010) and perform progress visualization (Machiraju et al. 2001). Generally, a saliency map is computed from the data using center-surround operation and Gaussian-weighted average of the emphasis field centered at a voxel (Itti et al. 1998). Our method also uses Gaussian-weighted center-surround operation in volume data attribute space.

While most research in image saliency only considers a single image at time, Jacobs et al. (2010) proposed a new concept cosaliency to explore the importance of local structure changes between images. In general, the cosaliency of an item, e.g., a voxel in the flow field, is its state or quality of standing out with respect to other flow field. A proper cosaliency model could help to visualize the most cosalient features that summarize the differences across an image set (Jacobs et al. 2010).

Kim et al. (2011) proposed a method to detect salient regions of images or videos. The method combines the spatial saliency by computing distances between ordinal signatures of edge and color orientations obtained from the center and the surrounding regions and the temporal saliency by simply computing the sum of absolute difference between temporal gradients of the center and the surrounding regions.

2.2 Transfer functions for time-varying data

Designing transfer functions for time-varying data is an important problem in volume visualization (Ayala et al. 2005). The early work called the contour spectrum (Bajaj et al. 1997) allows the user to display the changes of the underlying contour functions over time. The contour spectrum is a statistical signature consisting of a variety of scalar data and contour attributes, computed over the range of scalar values. Isovalues of interest can be chosen manually from different times and contour function values. The result is a set of opacity maps over time.

Jankun-Kelly and Ma (2001) studies different semi-automatic approaches for transfer function generation to obtain a single or minimal set of informative opacity transfer functions for a time-varying dataset. They propose two types of methods that obtain opacity functions: summary functions and summary volume. The first one consists of algorithms which analyze each frame separately, create a transfer function, and then try to combine these transfer functions into a summary function (in a similar way as 3D+time). Summary-volume-based techniques do not ignore the temporal dimension and operate upon the entire set of volumes to generate a transfer function. It creates a single volume that combines all frames and then they extract a transfer function from this volume.

Wang et al. (2011) presented a volume exploration scheme that explores volumetric structures in the feature space by modeling the space using the Gaussian mixture model (GMM). They extended GMM-based exploration scheme to time-varying datasets using an incremental GMM estimation algorithm. The computed elliptical transfer functions can be freely adjusted and the adjustments are then automatically propagated to other time steps.

Although transfer functions are designed to visualize volume data to human for further analysis, few methods have taken human perception into account and incorporate models of low-level human visual attention. We apply the defined spatiotemporal volume saliency to transfer functions so that prominent time-varying volume structures can be captured and visualized automatically.

3 Spatiotemporal volume saliency computation

Our spatiotemporal volume saliency computation integrates two parts: spatial and temporal saliency, which are discussed progressively in this section.

3.1 Spatial saliency

Kim and Varshney (2006) introduced saliency into interactive visualization to enhance volume rendering. They present a visual-saliency-based operator to enhance selected regions of a volume. The method starts with generation of a saliency field S that defines a value of saliency for every voxel. They used the center-surround mechanism (Itti et al. 1998) on the volume attribute. The saliency field S is computed with the Laplacian of Gaussian-weighted average of the attribute field centered at a voxel. This saliency field is mainly to highlight selected parts of volume datasets with obvious structure boundaries. We adapted the method to generate saliency volume for scientific data.

We verify the effectiveness of the adopted spatial volume saliency for volume with various datasets as shown in Fig. 1. These volume datasets are from different fields. The spatial saliency could automatically extract salient structures from the raw volume.

3.2 Temporal saliency

For time-varying volume data, the objective of temporal saliency is to reflect the dynamic nature of time-varying data. Therefore, the value changes of attributes decide whether the voxel is saliency or not, which is confirmed by the success of this idea on 2D problems (Kim et al. 2011). This has led us to formulate

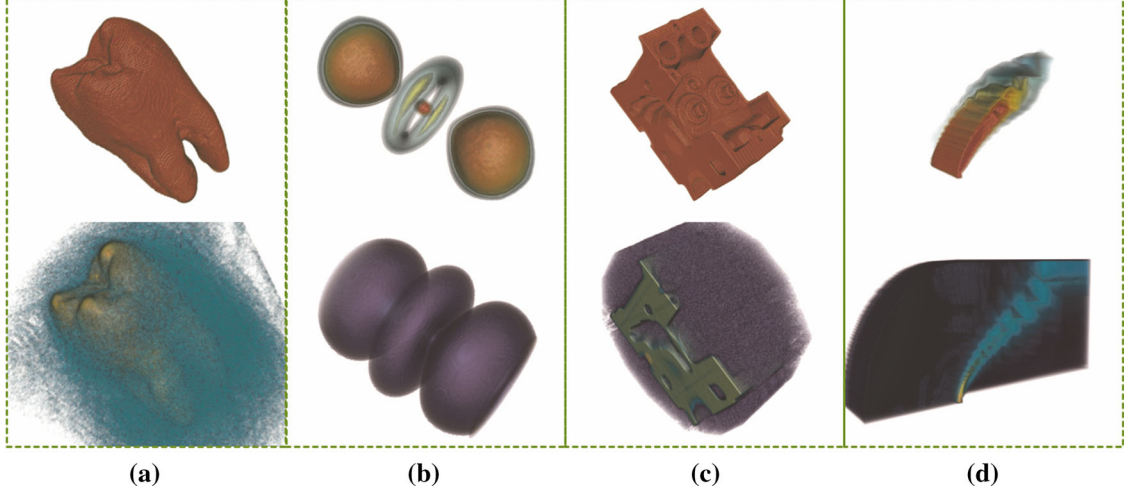


Fig. 1 Spatial saliency verification: **a** tooth, **b** hydrogen atom (H), **c** engine, and **d** bluntfin. The salient structures in the raw volume (the *bottom* row) are captured and visualized (the *top* row) automatically with the spatial saliency

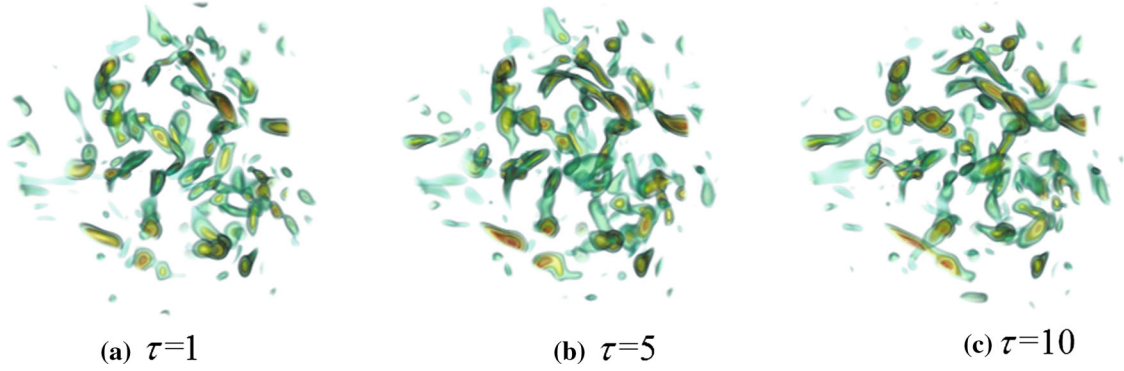


Fig. 2 Temporal saliency with different time intervals τ . One could find time-varying behaviors with different periods

temporal saliency in terms of the mean attribute gradient between steps used with the center-surround mechanism (Itti et al. 1998; Lee et al. 2005).

The first step of our temporal saliency computation involves computing attribute gradient between steps, namely temporal attribute gradient (TAG). Let $a_v(n)$ denote the attribute value of voxel v at time step n . The TAG $\mathcal{G}^T(v)$ is computed as

$$\mathcal{G}^T(v) = |a_v(n) - a_v(n - \tau)| \quad (1)$$

where τ indicates time interval. The positive time interval τ builds the temporal saliency of one step volume data based on the attribute gradient with its preceding step, while negative τ builds the temporal saliency of one step volume data based on the attribute gradient with its subsequent step. Different time intervals could focus on different behaviors of time-varying data. For example, if $\tau = 1.0$, the temporal saliency will further capture random/hot behaviors. If $\tau = 5.0$ or 10.0 , the temporal saliency will get time-varying behaviors with longer period. Figure 2 illustrates the results of the temporal saliency with different time intervals. In practice, the time interval τ is set to 1.0 generally.

Let the neighborhood $N(v, \sigma)$ for a voxel v be the set of voxel within a distance σ . We consider Euclidean distance as distance function. Thus, $N(v, \sigma) = \{x | \|x - v\| < \sigma, x \text{ is a voxel}\}$. Let $\mathcal{N}(\mathcal{G}^T(v), \sigma)$ denote the Gaussian-weighted average of the TAG. We compute this as

$$\mathcal{N}(\mathcal{G}^T(v), \sigma) = \frac{\sum_{x \in N(v, 2\sigma)} \mathcal{G}^T(x) \exp[-\|x - v\|^2 / (2\sigma^2)]}{\sum_{x \in N(v, 2\sigma)} \exp[-\|x - v\|^2 / (2\sigma^2)]}. \quad (2)$$

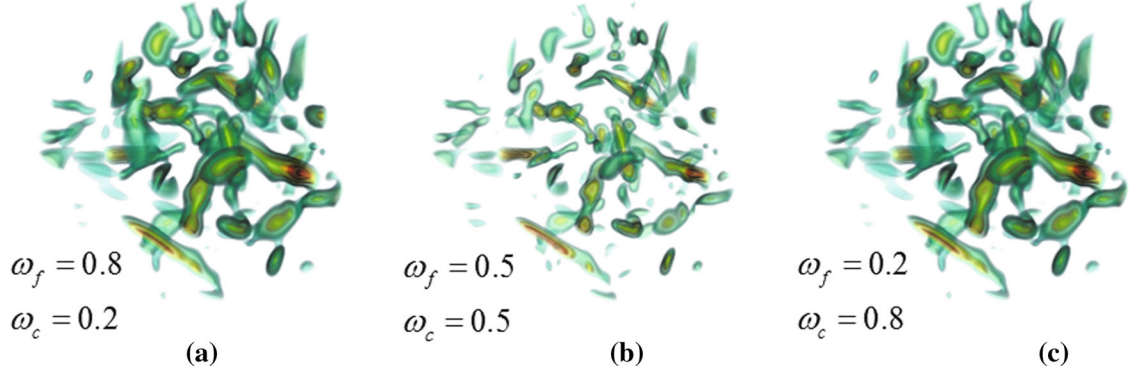


Fig. 3 Effects illustration of different weights of fine ω_f and coarse ω_c scales in the Gaussian-weighted averages

As in Lee et al. (2005), we assume a cut-off for the Gaussian filter at a distance 2σ . And the temporal saliency $\mathcal{S}^T$ of a voxel v is computed as the absolute difference between the Gaussian-weighted averages computed at fine and coarse scales:

$$\mathcal{S}^T(v) = |\omega_f \mathcal{N}(\mathcal{G}^T(v), \sigma) - \omega_c \mathcal{N}(\mathcal{G}^T(v), 2\sigma)| \quad (3)$$

where ω_f and ω_c denote weights of fine and coarse scales, respectively. Experiments (Fig. 3) found that the smaller the absolute difference between ω_f and ω_c , the more concentrate the temporal saliency.

To filter inconspicuous variation and highlight the development of important structures, the temporal saliency is computed at multiple scales:

$$\mathcal{S}_i^T(v) = |\omega_f \mathcal{N}(\mathcal{G}^T(v), \sigma_i) - \omega_c \mathcal{N}(\mathcal{G}^T(v), 2\sigma_i)| \quad (4)$$

where σ_i is the standard deviation of the Gaussian filter at scale i . In practice, we have used five scales $\sigma_i \in \{3\varepsilon, 4\varepsilon, 5\varepsilon, 6\varepsilon, 7\varepsilon\}$, where ε is the length of the volume grid.

To integrate temporal saliency $\mathcal{S}_i^T$ at different scales, we apply a non-linear suppression operator S similar to the one proposed by Itti et al. (1998). After normalizing temporal volume saliency $\mathcal{S}_i^T$ to a fixed range $[0, 1]$, we compute the average of all its other local maxima without global maxima $\bar{m}_i$ which is equal to 1.0 in practice. Then, we globally multiply the map by $(1.0 - \bar{m}_i)^2$. The result temporal saliency $\mathcal{S}^T$ is computed by adding the saliency maps at all scales after applying the non-linear normalization of suppression (Lee et al. 2005):

$$\mathcal{S}^T(v) = \sum_i \mathcal{S}_i^T(v) \quad (5)$$

Figure 4 lists the results with different scales σ_i (Fig. 4a–e) and the final temporal saliency (Fig. 4f). Larger scale is helpful to filter out little salience structures and reduce visual clutter. The integration of temporal saliency at different scales could accentuate visual foci and preserve certain contexts.

3.3 Integration

Finally, the proposed methodology for measuring the spatiotemporal volume saliency relies on the weighted combination of the spatial and temporal saliencies:

$$\mathcal{S}(v) = (1 - \omega_t) \mathcal{S}^C(v) + \omega_t \mathcal{S}^T(v) \quad (6)$$

where

$$\mathcal{S}^C(v) = \sum_i (|\omega_f \mathcal{N}(a(v), \sigma_i) - \omega_c \mathcal{N}(a(v), 2\sigma_i)|) \quad (7)$$

$a(v)$ and $\omega_t \in [0, 1]$ denote the attribute value of voxel v and the weighting factor for balancing between the spatial and temporal saliency, respectively. One could modify ω_t to get different saliency that focuses on spatial or temporal saliency. For example, as shown in Fig. 5, $\omega_t = 0.3$ will focus more on spatial saliency,

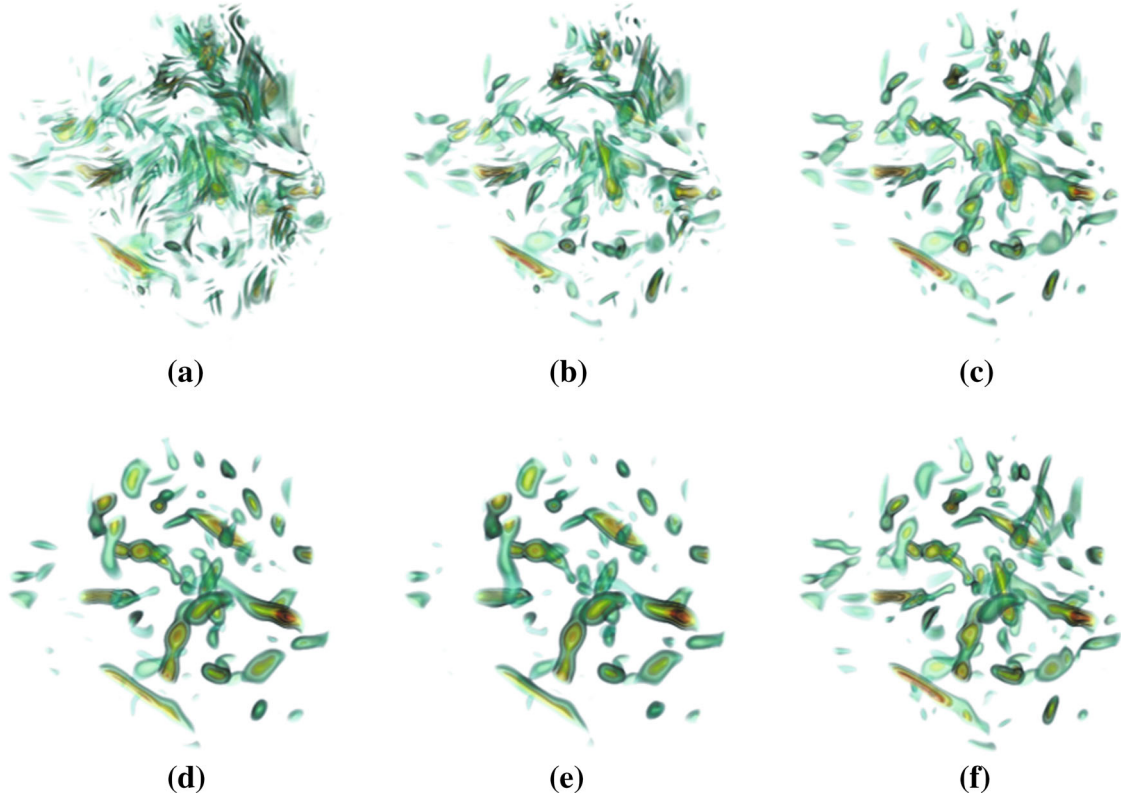


Fig. 4 **a–e** The saliency at scales of 3ε , 4ε , 5ε , 6ε , and 7ε . **f** The final temporal saliency after aggregating the saliency over multiple scales. Here, ε is the length of the volume grid

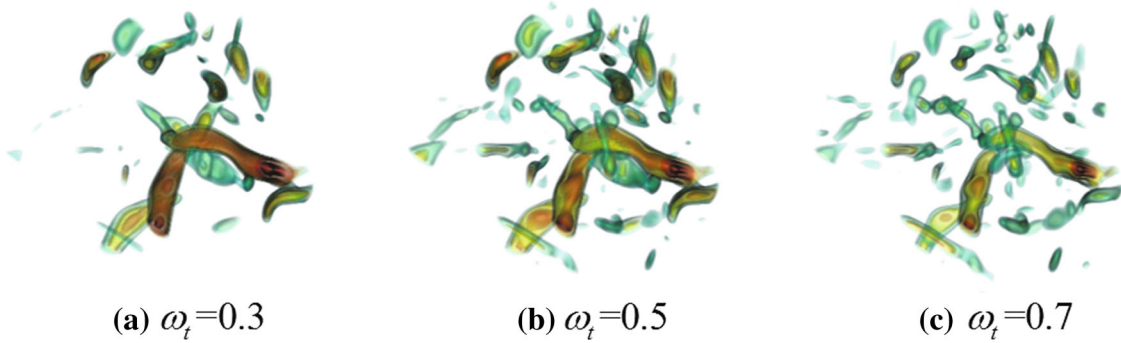


Fig. 5 Different weights of temporal saliency ω_t . **a** A low value of ω_t like 0.3 will focus more on the spatial saliency while $\omega_t = 0.5$ provides the balanced saliencies **(b)**. **c** A high value of ω_t like 0.7 will focus more on the temporal saliency

while $\omega_t = 0.7$ will focus more on temporal saliency. $\omega_t = 0.5$ provides the balanced saliencies, which has been used in our practice.

3.4 Implementation details

As discussed, the spatiotemporal volume saliency is defined on voxel. We speedup the computation of the saliency by GPU with CUDA. We divide the computation of the saliency of different voxels into different threads on GPU. In practice, we use GPU-accelerated volume ray casting to render the data.

4 Application

4.1 Salient automatic transfer functions design

One of the main challenges for effective time-varying volume visualization is transfer function design. We applied the spatiotemporal volume saliency to generate a simple but effective transfer functions for time-varying volume data automatically (Fig. 6). The saliency was used to compute the opacity of voxel v . The color was mapped based on the attribute value.

The spatiotemporal volume saliency-based opacity computation begins with the statistics of the histogram distribution H_S of saliency. We use the least square function $f(H_S)$ to fit the histogram distribution H_S . Let s_l be the largest value that $f'(s_l) = 0$ if there is, or $s_l = \max(H_S)$. The threshold value $c(H_S)$:

$$c(H_S) = s_i \quad \text{if } f'(s_i) = \omega_S \text{ and } s_i \leq s_l \quad (8)$$

where ω_S is a user-defined threshold of the first-order derivative of the $f(H_S)$. Let N be the normalization function. The result opacity function is

$$o(v_j) = \begin{cases} N(s_j - c(H_S)) & \text{if } s_j > s_i \\ 0.0 & \text{else.} \end{cases} \quad (9)$$

The range of focused region can be controlled with ω_S . In practice, we set $\omega_S = 0.3$. For example, if the histogram distribution is bipolar which is common in the volume datasets with explicit structure boundaries, like Fig. 6a, only the prominent structures are rendered (Fig. 6a–c) with our opacity function. If the volume data have no explicit structure boundary, the histogram distribution is somewhat similar with normal distribution function, like Fig. 6d, only the prominent features are rendered (Fig. 6f). In Fig. 6g, as the context changed greatly, the rendering result with our opacity function includes the dramatically varying context besides important features (Fig. 6i).

4.2 Salient key frame selection

Time-varying datasets always consist of hundreds of steps. It is helpful to automatically detect some key frames in exploration process. Wang et al. (2008) presented an importance-driven approach to select time steps based on information theory. We integrate information theory (Wang and Shen 2011) with the spatiotemporal volume saliency and propose Saliency Entropy to measure importance H_S of step t_i relative to step t_j :

$$H_S(t_i|t_j) = H_S(t_i) - I_S(t_i; t_j) \quad (10)$$

where

$$I_S(t_i; t_j) = \sum_{x \in t_i} \sum_{y \in t_j} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (11)$$

To evaluate the importance of step t_i , we first calculate its saliency entropy $H_S(t_i)$ and then compute its mutual information $I_S(t_i; t_j)$ with step t_j . These quantities are used to derive the importance of step t_i using conditional saliency entropy $H_S(t_i|t_j)$.

Figure 7a shows the result saliency entropy with the turbulent vortex dataset. One could easily find that time steps 11 and 60 (Fig. 7b, c) with large saliency entropy (in red) are remarkably different with other steps. Time steps 0, 2, 18 and 32 with low saliency entropy are similar with each other in spatiotemporal volume saliency. Different from the discrete distributed saliency entropy of the turbulent vortex dataset, the saliency entropy with the turbulent combustion simulation dataset is successive, as shown in Fig. 8a. The blue pixel highlighted with black rectangle illustrates that the steps 48 and 66 (Fig. 8b, c) are similar in the view of spatiotemporal volume saliency. Steps 45, 60, 86 and 101 (Fig. 8e–g) are obviously different with the other steps.

5 Results

In this section, we test and verify of the proposed spatiotemporal volume saliency in detail.

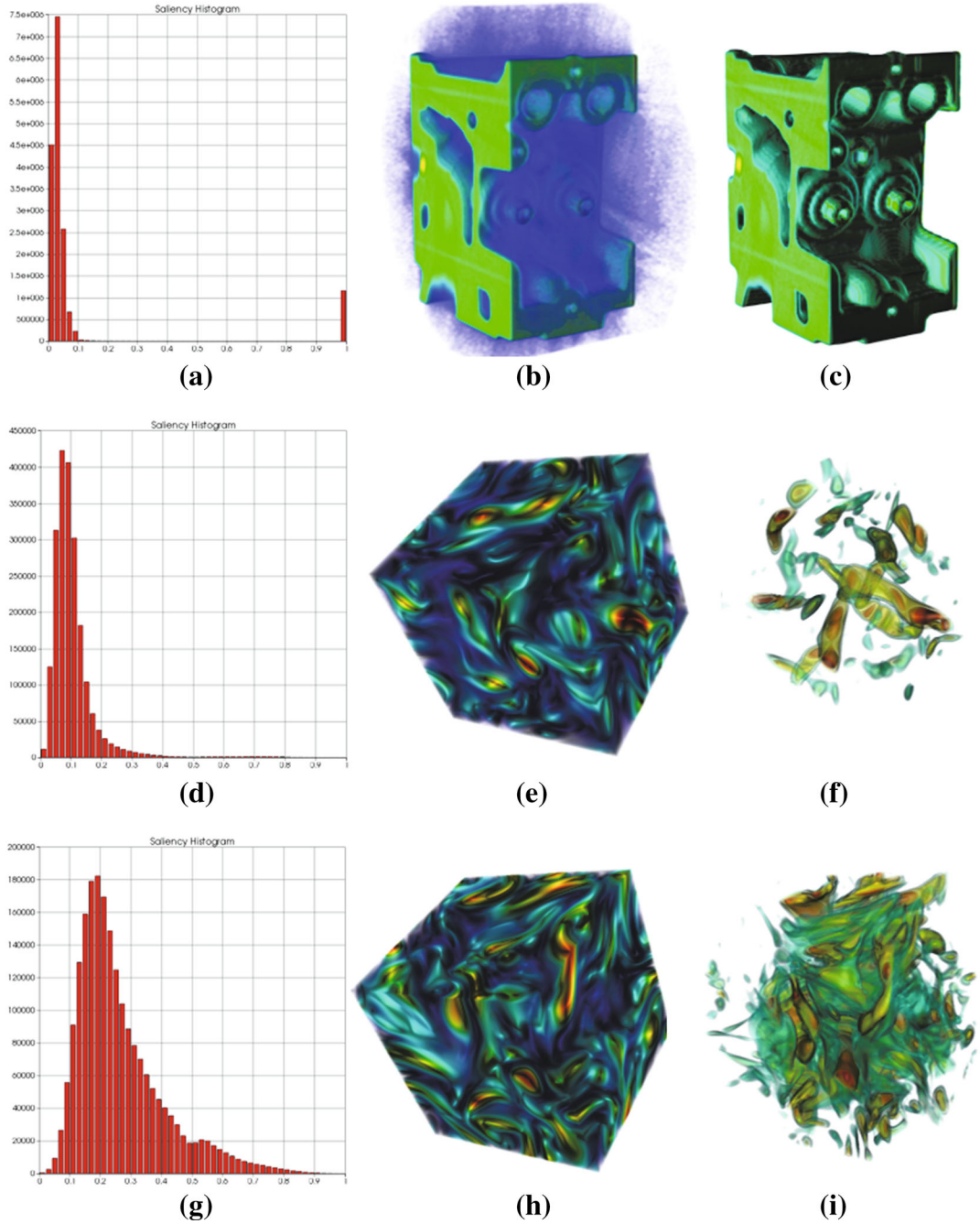


Fig. 6 Salient automatic transfer functions design. *Left column* histogram distribution of spatiotemporal volume saliency. *Middle column* images with transfer function based on attribute. *Right column* results from the transfer function based on spatiotemporal volume saliency

5.1 Datasets and performance

We tested our spatiotemporal volume saliency with two kinds of volume datasets, static and time-varying. As discussed in Sect. 3.1, the static volume datasets are mainly used to confirm the effectiveness of the adapted saliency filter in volume. Their configurations are listed in Table 1 and the results are shown in Fig. 1. The time-varying volume datasets are listed in Table 1. The values of the defined parameters were set empirically. But we found that these values are efficient for the tested datasets throughout.

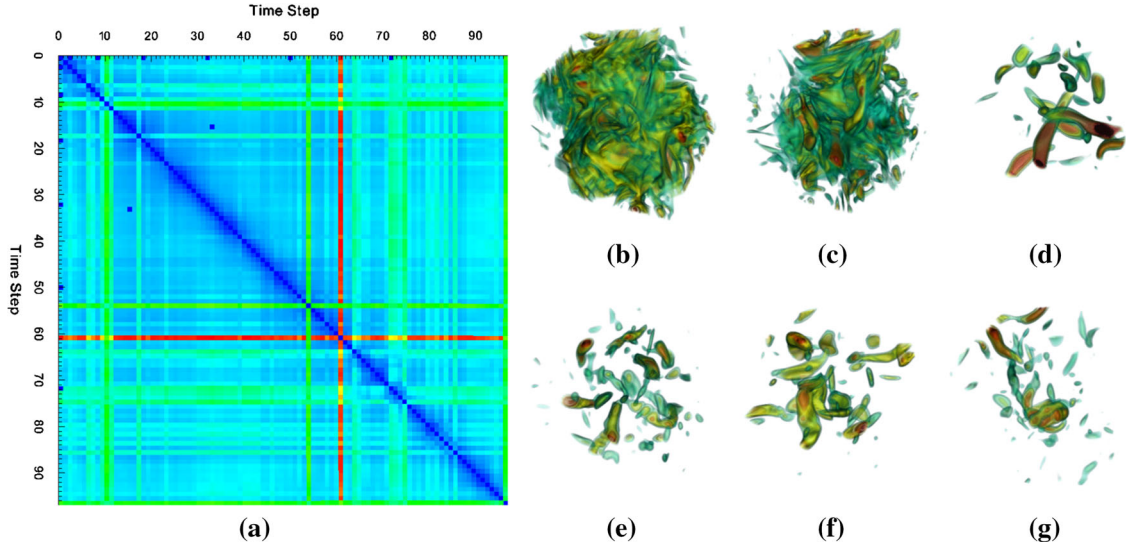


Fig. 7 Salient key frame selection with the turbulent vortex dataset: **a** the saliency entropy field. Warmer colors (reds and yellows) show high saliency entropy and cooler colors (greens and blues) show low saliency entropy. **b, c** Images of time step 11 and 60 which have large saliency entropy. **d–g** Images of time step 0, 2, 18 and 32 with similar saliency and low mutual saliency entropy

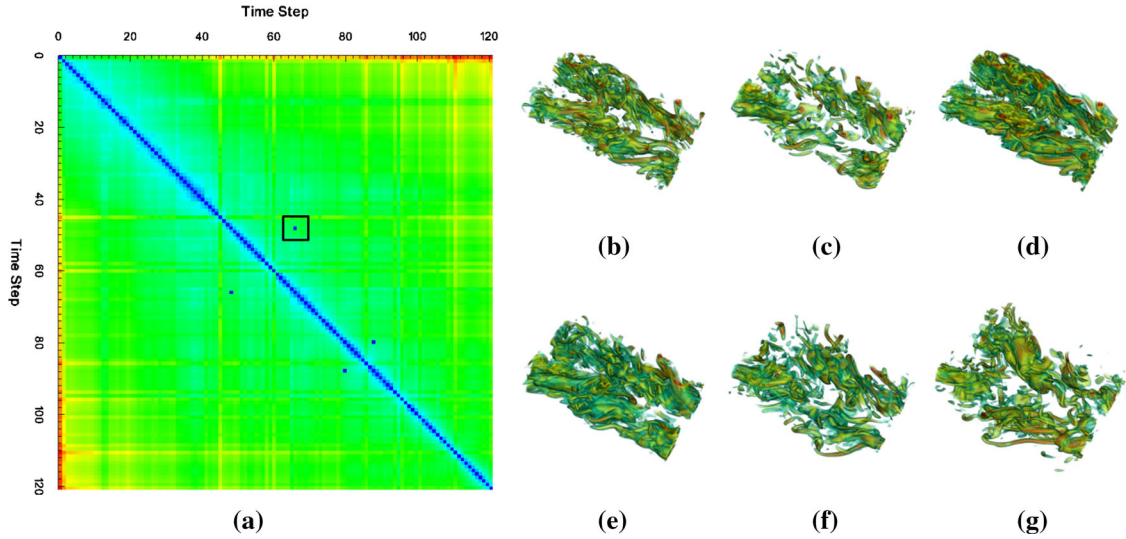


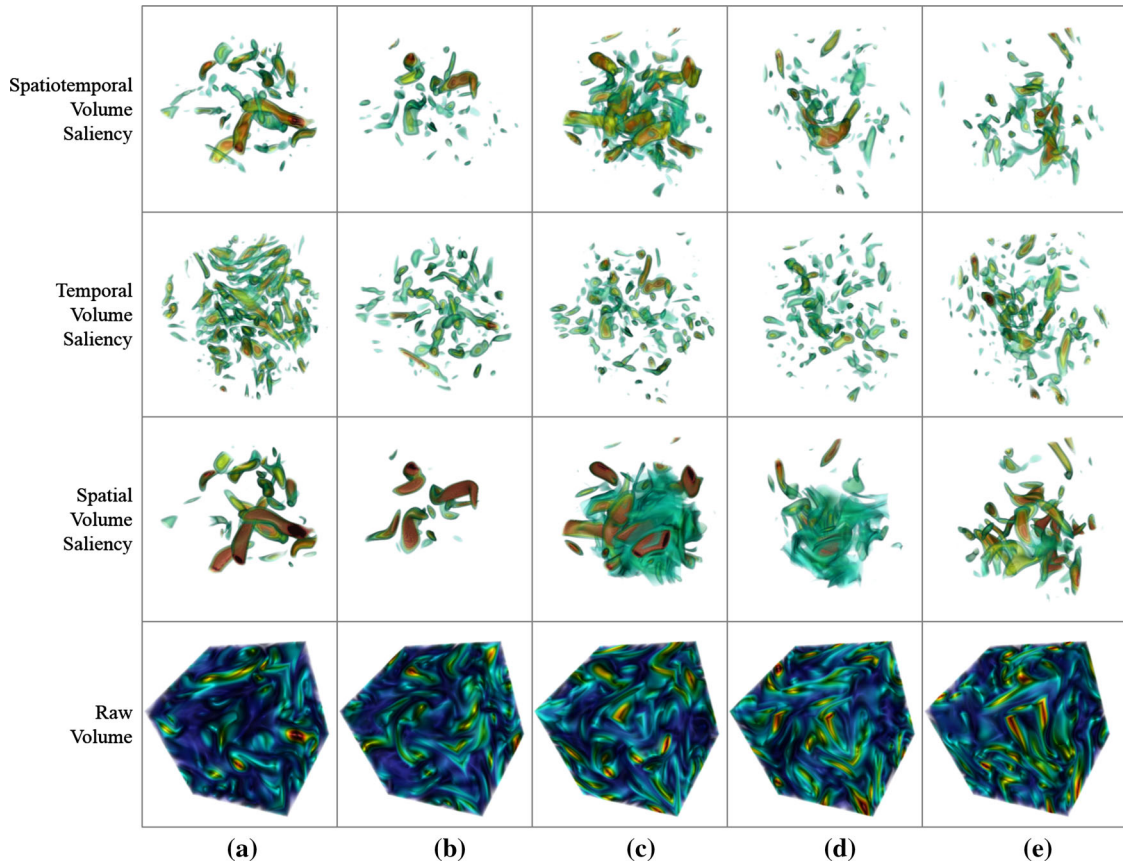
Fig. 8 Salient key frame selection with the turbulent combustion simulation dataset: **a** the saliency entropy field. Warmer colors (reds and yellows) show high saliency entropy and cooler colors (greens and blues) show low saliency entropy. **b, c** Images of time steps 48 and 66 which have low saliency entropy are similar in the view of spatiotemporal volume saliency. **d–g** Images of time steps 45, 60, 86 and 101 are obviously different with other steps

Although the spatial volume saliency (Kim and Varshney 2006) is efficient for small volume, it is not efficient enough for large time-varying volume data. We deal with this problem by migrating this method into GPU and get the speedup of nearly 150 times. Similarly, we also implemented temporal saliency computation with CUDA and got the speedup of about 110 times.

The performance of the proposed method was tested on a desktop computer, with a quad-core Intel i7-950 2.8 GHz processor with 12 GB memory and an Nvidia Geforce K5000 graphics card with 4096 MB video memory. The timings of the datasets are listed in Table 1. The computation of spatial and temporal saliency was performed using CUDA 6.0.

Table 1 The volume datasets tested in our experiments, their configurations and processing timing performance in seconds

Dataset	Dimension	Time step	Spatial saliency (s)	Temporal saliency (s)	Total (s)
Tooth	$256 \times 256 \times 161$	—	0.12	—	0.12
H	$128 \times 128 \times 128$	—	0.02	—	0.02
Engine	$256 \times 256 \times 256$	—	0.24	—	0.24
Bluntfin	$256 \times 128 \times 064$	—	0.02	—	0.02
Turbulence	$128 \times 128 \times 128$	99	0.02	0.10	0.12
Combustion	$240 \times 360 \times 060$	122	0.09	0.53	0.62

**Fig. 9** Spatiotemporal volume saliency of the turbulence vortex dataset (99 time steps): the spatiotemporal volume saliency (*top row*) which combines spatial and temporal volume saliency (*third and second rows from the top*) effectively captures the prominent time-varying structures (vortex). We take snapshots of the time-varying data at steps 1, 20, 39, 58 and 77 from *left to right*

5.2 Time-varying volume results

Figure 9 shows the results with a turbulence vortex simulation dataset. The fourth row shows the raw volume dataset. The third and second rows show the spatial and temporal saliency, respectively. The first row shows the result spatiotemporal volume saliency results. Take the first column for example, the spatial saliency extracts the prominent structures in the step 1, while the temporal saliency captures the notable variations comparing with step 0. Such results are not available in other time-varying volume visualization work using the same dataset. Moreover, notice how the first row in Fig. 9 succinctly depict the volume patterns using a small part of the raw data. It is necessary to notice that the spatiotemporal volume saliency could reveal important time-varying structures that are not available for the spatial volume saliency, which is introduced from the exiting work (Kim and Varshney 2006). Please refer to our attached video to view the whole dynamic process with spatiotemporal volume saliency.

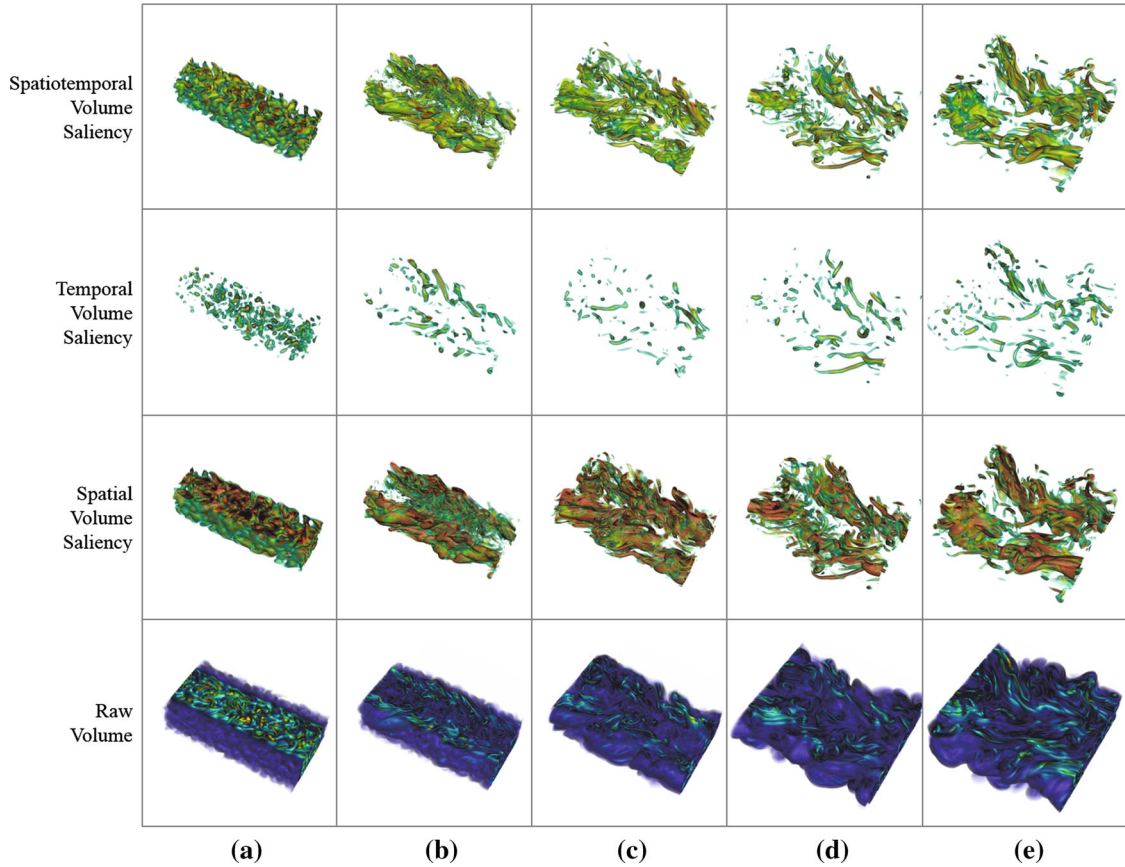


Fig. 10 Illustration of spatiotemporal volume saliency by turbulence combustion simulation dataset (122 time steps): we take snapshots of the time-varying data at steps 1, 31, 61, 91 and 121 from *left to right*. Optionally, the user could choose to observe salient structures in each step (the *third row*) or regions with makable variation only (the *second row*)

Besides the turbulence vortex simulation dataset, we also experimented with a turbulent combustion volume. In Fig. 10, we show the raw volume, spatial and temporal saliency and the result spatiotemporal volume saliency from the bottom to up. In Fig. 10a, the spatiotemporal volume saliency successfully revealed the prominent structures inside the combustion at the beginning. As the development of the combustion process, the noticeable structures spread to the sides of the whole volume. These results further verified the utility of spatiotemporal volume saliency in time-varying volume datasets. Optionally, the user could choose to observe salient structures in each step or region with makable variation only.

6 Conclusions and future work

Effective visualization of time-varying volume fields remains a challenge, which is exacerbated by the increase in size and complexity of volume datasets ever produced. The spatiotemporal volume saliency we have introduced offers a new way to characterize and visualize the volume structure and patterns. The spatiotemporal volume saliency highlights prominent features, and their development reveals the remarkable variation in the whole volume and reduces visual occlusion greatly.

We have applied the proposed spatiotemporal volume saliency to transfer function design and key frame selection. Next, we will further compare the results with exiting techniques. It is feasible to make dynamic view selection for time-varying volumes based on spatiotemporal volume saliency. After selecting key frames, the views for these key frames can be computed based on spatiotemporal volume saliency. Then the path of views between key frames can be interpolated under certain rules (Ji and Shen 2006). In the future, we will also apply our technique to other real-world volume datasets and involve domain scientists in the evaluation of this new approach.

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